Extensors (in french: ‘Extenseurs’)

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Abstract

For now, this article is a mere translation of this work: `http://www.numdam.org/item/DIA_1980__3__A3_0.pdf`. The hope is to expand this quite laconic article with details.

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0 Preamble (for me)

[This section is important for me. It may be deleted in the final version. Let me recall some words here, and then (try to) set a strategy to attack the paper.]

A *monad* for a category $\mathcal{C}$ consists of:

- one functor $T : \mathcal{C} \to \mathcal{C}$
- one natural transformation $\eta : 1_{\mathcal{C}} \to T$
- one natural transformation $\mu : T^2 \to T$, where $T^2 := T \circ T$

such that the diagrams

$$\begin{tikzcd} T^3x \arrow[r, "T\mu_x"] \arrow[d, "\mu_{Tx}"'] & T^2x \arrow[d, "\mu_x"] \\ T^2x \arrow[r, "\mu_x"'] & Tx \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} Tx \arrow[r, "\eta_{Tx}"] \arrow[d, "T\mu_x"'] \arrow[dr, "1_{Tx}"] & T^2x \arrow[d, "\mu_x"] \\ T^2x \arrow[r, "\mu_x"'] & Tx \end{tikzcd}$$

commute for every object x of $\mathcal{C}$; let us write such monad as the triple (T, η, μ).

In the sketched proof, the author talks about some ‘monade des parties du topos’. **Set** is a topos and

Sandbox 0.1 (The ‘power set’ monad)**.** Consider the functor $2^\bullet : \mathbf{Set} \to \mathbf{Set}$ where, for X a set, $2^\bullet(X) := 2^X$ and, for $f : X \to Y$, define $2^f := 2^\bullet(f)$ to be the function $2^A \to 2^B$ that takes subsets E of A to fE.
Now, we need functions $\xi_X : X \to 2^X$, one for each set X, forming a natural transformation $\xi : 1_{\mathbf{Set}} \to 2^\bullet$; we consider that one introduced as $\xi_X(x) := \{x\}$.
To conclude the structure of monad, consider for X set, the function

$$\delta_X : 2^{2^X} \to 2^X\,,\ \delta_X(F) := \bigcup_{E \in F} E.$$

The family of such functions determine a natural transformation $\delta : 2^\bullet \circ 2^\bullet \to 2^\bullet$.

Let us try now to bring this to any topoi, if possible.

Sandbox 0.2 (The monad of parts for a topos $\mathcal{E}$, attempt #1)**.** One of the possible definitions of (elementary) topos is:

a category with finite products and power objects.

For future reference, here is the definition of power object:

For if $\mathcal{C}$ is a category with finite products and $a \in |\mathcal{C}|$, a *power object* in $\mathcal{C}$ consists of

- one object, 'of the parts of a', we write Ω^a
- one monomorphism $\in_a \to \Omega^a \times a$, the 'membership relation'

  with the property: for every $b \in |\mathcal{C}|$ and monomorphism $r \to b \times a$ there exists one and only one morphism $f : b \to \Omega^a$ for which there is a pullback square

$$\begin{tikzcd} r \arrow[r] \arrow[d] & b \times a \arrow[d, "f \times 1_a"] \\ \in_a \arrow[r] & \Omega^a \times a \end{tikzcd}$$

  in $\mathcal{C}$.

Now if we want to replicate the above phenomenon inside any topos $\mathcal{E}$, we need a suitable functor $\Omega^\bullet : \mathcal{E} \to \mathcal{E}$. For $a \in |\mathcal{E}|$,

$$\Omega^\bullet(a) := \Omega^a.$$

It remains to understand what $\Omega^f := \Omega^\bullet(f)$, for $f : a \to b$ in $\mathcal{E}$, could be to make a functor. Consider the power objects of a and b, that is Ω^a and Ω^b together with their respective membership relations $\in_a \to \Omega^a \times a$ and $\in_b \to \Omega^b \times b$.

$$\begin{tikzcd} \in_a \arrow[r] & \Omega^a \times a \arrow[r, "1_{\Omega^a} \times f"] & \Omega^a \times b \arrow[d, dashed] \\ \in_b \arrow[rr] & & \Omega^b \times b \end{tikzcd}$$

Here, the dotted morphism is of the form $\Omega^f \times 1_b$, where $\Omega^f : \Omega^a \to \Omega^b$ that one for which there is a pullback square

$$\begin{tikzcd} \in_a \arrow[r] \arrow[d] & \Omega^a \times a \arrow[r, "1_{\Omega^a} \times f"] & \Omega^a \times b \arrow[d, dashed] \\ \in_b \arrow[rr] & & \Omega^b \times b \end{tikzcd}$$

Let us verify functoriality now. Take

$$a \xrightarrow{f} b \xrightarrow{g} c$$

so that we have also

$$\Omega^a \xrightarrow{\Omega^f} \Omega^b \xrightarrow{\Omega^g} \Omega^c.$$

Let us study $\Omega^{gf} : a \to c$. By definition, Ω^{gf} is the morphism for which there is a pullback square

$$\begin{tikzcd} \in_a \arrow[r] \arrow[d] & \Omega^a \times a \arrow[r, "1_{\Omega^a} \times (gf)"] & \Omega^a \times c \arrow[d, "\Omega^{gf}"] \\ \in_c \arrow[rr] & & \Omega^c \times c \end{tikzcd}$$

We have then two pullback squares glued like this:

$$\begin{tikzcd}
\in_a \arrow[r] \arrow[d] & \Omega^a \times a \arrow[r, "1_{\Omega^a} \times f"] & \Omega^a \times b \arrow[d, "\Omega^f \times 1_b"'] & \\
\in_b \arrow[rr] \arrow[d] & & \Omega^b \times b \arrow[r, "1_{\Omega^b} \times g"'] & \Omega^b \times c \arrow[d, "\Omega^g \times 1_c"] \\
\in_c \arrow[rrr] & & & \Omega^c \times c
\end{tikzcd}$$

The first one gives the morphism Ω^f, whereas the second one Ω^g. Let us add two arrows to make a rectangle:

$$\begin{tikzcd}
\in_a \arrow[r] \arrow[d] & \Omega^a \times a \arrow[r, "1_{\Omega^a} \times f"] & \Omega^a \times b \arrow[r, "1_{\Omega^a} \times g"] \arrow[d, "\Omega^f \times 1_b"'] & \Omega^a \times c \arrow[d, dashed] & \\
\in_b \arrow[rr] \arrow[d] & & \Omega^b \times b \arrow[r, "1_{\Omega^b} \times g"'] & \Omega^b \times c \arrow[d, "\Omega^g \times 1_c"] \\
\in_c \arrow[rrr] & & & \Omega^c \times c
\end{tikzcd}$$

Here, the dotted arrow is the unique that could be, that is $\Omega^f \times 1_c$; this results from the pact that in the upside rectangle on the right, the morphisms involved are all monomorphisms. The composition of the morphisms $\Omega^a \times c \to \Omega^b \times c \to \Omega^c \times c$ on the right is $(\Omega^g \Omega^f) \times 1_c$: this means that if we can show the perimetric rectangle is a pullback square [Pullback Lemma here...], we can conclude $\Omega^{gf} = \Omega^g \Omega^f$. [TEX that!] [There is a problem here... fixing it...]

[Examples here! Or not exactly here, but more examples for myself...]

If you are given a monad (T, η, μ) for a category $\mathcal{C}$, you can introduce an *algebra* for this monad, consisting of:

- one object x in $\mathcal{C}$
- one morphism $f : Tx \to x$

all this making commute the following diagrams

$$\begin{tikzcd}
T^2 x \arrow[r, "Tf"] \arrow[d, "\mu_x"'] & Tx \arrow[d, "f"] \\
Tx \arrow[r, "f"'] & x
\end{tikzcd}
\quad \text{and} \quad
\begin{tikzcd}
x \arrow[dr, "\eta_x"] \arrow[dd, "1_x"'] & \\
& Tx \arrow[dl, "f"] \\
x &
\end{tikzcd}$$

[I've just written a couple of definitions, but for the next days here is what to do for me: find more examples motivating monads and algebras for monads; you must learn something about *Kleisli* and *Eilenberg-Moore categories*... Not only for the sake of these pages.]

[Now understand this: 'un treillis complet est une $2^\bullet$-algèbre'.]

1 The actual article

[The symbolism here is weird...]

An *extensor* [in french, it is 'extenseur'] for a category $\mathcal{C}$ amounts at having one object a of $\mathcal{C}$ and for every *span*

$$\begin{tikzcd}
& x \arrow[dl, "f"'] \arrow[dr, "g"] & \\
a & & y
\end{tikzcd}$$

of $\mathcal{C}$ one morphism $y \to a$ of $\mathcal{C}$ we denote $\langle f, g \rangle$; all this complying with the following rules:

1. For any consecutive morphisms $x \xrightarrow{g} y \xrightarrow{h} z$ we have

$$\langle \langle f, g \rangle, h \rangle = \langle f, hg \rangle .$$

[Here is a drawing of all this:

$$\begin{tikzcd}
 & x \arrow[dl, "f"'] \arrow[dr, "g"] & \\
a & & y \arrow[ll, dashed, "\langle f,g \rangle"'] \arrow[d, "h"] \\
 & & z \arrow[ull, dotted, "\langle \langle f,g \rangle,h \rangle = \langle f,hg \rangle"]
\end{tikzcd}$$

]

2. If

$$\begin{tikzcd}
 & \arrow[r, "p'"] \arrow[d, "h'"'] & \arrow[d, "h"] \\
 & \arrow[r, "p"'] &
\end{tikzcd}$$

is a pullback square [in french, it is 'produit fibré', that is fibred product], then for every f as in

$$\begin{tikzcd}
 & \arrow[r, "p'"] \arrow[d, "h'"] & \arrow[d, "h"] \\
 & \arrow[r, "p"'] & \\
\arrow[ur, "f"] & &
\end{tikzcd}$$

we have

$$\langle f, p \rangle h = \langle fh', p' \rangle .$$

3. If in

$$\begin{tikzcd}
 & \arrow[l, "p"'] \arrow[r, bend left, "h"] & \arrow[l, bend left, "h'"] \arrow[r, "q"] &
\end{tikzcd}$$

we have $ph'h = p$ and $qhh' = q$, then for if f is as in

$$\begin{tikzcd}
 & \arrow[l, "f"'] & \arrow[l, "p"'] \arrow[r, bend left, "h"] & \arrow[l, bend left, "h'"] \arrow[r, "q"] &
\end{tikzcd}$$

then

$$\langle aph', q \rangle = \langle ap, qh \rangle .$$

[This definition is a mess.] [Rewrite?]

[TEX the unique result of the paper. What is a 'sup-treillis'? A 'treillis' is a lattice. Maybe a lattice with products?]

2 Proof

[Write the translation...][Yet to be TEX-ed...]